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Real-Time Visual SLAM Trajectory Optimization and Error Correction Technology under Multi-Scale Semantic Constraints

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Abstract: This article proposes a real-time SLAM framework that integrates pixel level, object level, and scene level semantic information to address the issue of traditional visual SLAM being susceptible to cumulative errors and semantic blind spots in dynamic and complex environments. This framework collaboratively optimizes trajectory optimization and error correction through multi-scale semantic constraints. The system constructs a lightweight multi branch semantic segmentation network to generate a unified semantic graph, and embeds it into the entire process of front-end tracking, keyframe selection, and back-end pose graph optimization, achieving semantic driven trajectory generation and residual feedback correction. Experiments on the TUM RGB-D, KITTI, and self collected Urban Campus-2025 datasets show that the proposed method significantly outperforms existing mainstream systems in dynamic interference and repetitive structure scenarios: ATE (absolute trajectory error) is as low as 0.112 meters, RPE (relative pose error) is 0.021 meters, semantic annotation IoU reaches 78.4%, and single frame processing delay is controlled at 35.2 milliseconds. Especially during long-term operation, the system effectively suppresses drift through semantic closed-loop, and the final ATE of the 45 minute sequence is only 0.103 meters. The results validated the effectiveness of multi-scale semantic constraints in improving localization accuracy, trajectory rationality, and system robustness, providing a feasible technical path for highly reliable visual navigation.

Keywords: Multi-scale semantic constraints; Visual SLAM; Trajectory optimization; Error correction

1 Introduction

Simultaneous Localization and Mapping (SLAM), as the basic technology of intelligent mobile system to realize environment awareness and autonomous navigation, has been widely used in robotics, augmented reality and autonomous driving in recent years [1]. In the complex dynamic environment, it is the core challenge of autonomous navigation of intelligent mobile system to realize high precision and low delay synchronous positioning and map construction [2]. Traditional visual SLAM methods mainly rely on geometric consistency constraints, estimate camera pose and reconstruct sparse or dense maps through feature point matching or photometric error minimization. Although ORB-SLAM, DSO and other systems have made remarkable achievements in the structured static environment, they still face severe challenges in the real dynamic scene [3,4]. On the one hand, illumination change, texture loss or repeated structure easily lead to feature mismatch and local trajectory distortion [5]. On the other hand, the frequent appearance of dynamic objects destroys the hypothesis of static world, resulting in the drift of pose estimation and even the failure of tracking [6]. Geometric constraints themselves lack the ability to understand the semantic structure of the scene, which makes it difficult for the system to distinguish the passable area from obstacles, to identify the mismatching caused by repeated structures, and to be easily affected by accumulated errors in long-term operation, resulting in trajectory drift and even map collapse [7].

In order to break through the bottleneck, researchers gradually introduce semantic information to enhance the cognitive ability of SLAM system [8]. In recent years, with the development of deep learning and semantic segmentation technology, semantic SLAM has gradually become a research hotspot, and the cognitive ability of the system to the environment is enhanced by introducing high-level semantic

information [9]. However, most of the existing semantic SLAM methods only use single-scale semantic tags, which can not effectively model multi-granularity semantic hierarchy from pixel details to scene layout [10]. This leads to a very loose coupling between semantic apriori and geometric optimization [11]. Therefore, it is difficult for them to achieve accurate path guidance and error suppression under real-time constraints [12,13]. At present, there is an urgent need for a new SLAM framework, which can not only deeply integrate multi-granularity semantic priors, but also take into account real-time reasoning and engineering deployment, so as to realize high-precision, robust and semantically reasonable autonomous navigation in complex dynamic environments [14,15].

In response to the above issues, this article proposes a real-time visual SLAM trajectory optimization and error correction technology framework under multi-scale semantic constraints. This framework no longer regards semantic information as an auxiliary post-processing module, but deeply embeds it into the entire chain of front-end tracking and back-end optimization. The system first extracts pixel-level semantic masks, object-level instance boundaries, and scene-level topological structures through a lightweight multi-scale semantic segmentation network. Then, it constructs a semantic-geometric joint state space, introducing cross-scale consistency constraints in keyframe selection, feature matching, and pose graph optimization. At the trajectory optimization level, the system uses a semantic topology map to identify traversable areas and semantic stable nodes, dynamically generating navigation trajectories that align with environmental semantic logic. At the error correction level, an adaptive correction mechanism based on semantic residuals is designed to perform online compensation for local pose distortions caused by dynamic object occlusions or weak-texture regions, while global consistency is reinforced through semantic loop closure detection. The entire system enhances localization accuracy and map integrity in complex urban scenes and indoor dynamic environments while ensuring real-time inference speed (≥ 25 FPS).

The technical contributions of this article are mainly reflected in three aspects:

First, a unified representation model integrating pixel-, object-, and scene-level semantics is constructed, overcoming the information limitations of traditional single-scale semantic labels.

Second, a semantics-driven trajectory optimization strategy is proposed, ensuring that the estimated camera trajectory not only satisfies geometric consistency but also aligns with environmental semantic rationality.

Third, a multi-scale semantic residual feedback mechanism is designed to achieve collaborative correction of local and global errors, effectively suppressing drift effects during long-term operation.

The experimental section validates the superiority of the proposed method on metrics such as ATE, RPE, and semantic consistency using public datasets and self-collected real-world scenes, and conducts comparative analyses with mainstream SLAM systems. These results demonstrate its robustness advantages in environments with strong dynamic interference and high structural repetitiveness.

2 Related Work

As the core support of intelligent system environment perception and autonomous navigation, visual SLAM technology has made progress in theoretical innovation and engineering application in recent years. Hong et al. concluded that in the fields of robotics, AR, etc., the geometric constraint method in SLAM technology is mature in static structured environment, but there are still problems of accumulated errors and semantic blind spots in complex dynamic scenes [16]. Early semantic SLAM research mostly used loose coupling strategy, and the semantic segmentation results were attached to the traditional system as post-processing modules. Hempel and Al-Hamadi proposed an online semantic mapping system, which preliminarily verified the

enhancement of high-level semantics to environmental understanding by fusing semantic tags and geometric maps in real time [17]. Leroy et al. designed a lightweight semantic segmentation strategy for embedded platforms, which ensured real-time performance and provided reliable pixel-level semantic priors for SLAM systems [18]. This kind of work lays the foundation for semantic information embedding front-end tracking, but it has not solved the problem of multi-scale semantic collaborative modeling.

In the aspect of dynamic environment adaptability, researchers put forward a variety of semantic-guided dynamic object processing mechanisms. Liu and RDMO-SLAM developed by Miura combine optical flow prediction and semantic tagging to achieve robust feature tracking in dynamic scenes [19]. Huang et al. further proposed ADM-SLAM, which cooperated with Deeplapv3+semantic segmentation through adaptive feature point extraction to maintain high-precision pose estimation under dynamic interference [20]. He et al. fused K-Means clustering with semantic information to construct DSK-SLAM system to distinguish static background from moving objects [21]. DopeSLAM developed by Roch et al. realizes high-precision semantic 3D reconstruction under the framework of ROS [22]. Islam et al. improve the stability of semantic RGB-D SLAM in dynamic environment by enhancing multi-view geometric constraints [23]. Li et al. combined semantic stereo vision with ORB-SLAM2, which can meet the positioning challenge of outdoor dynamic scenes [24].

The introduction of deep learning technology has spawned a new paradigm of end-to-end SLAM. SVR-Net proposed by Lang et al. adopts sparse voxel cyclic network to realize direct joint optimization of monocular SLAM and TSDF maps [25]. Zhang et al. explored the application of unsupervised deep learning in the pose correction of binocular vision odometer, and provided new ideas for error compensation [26]. Elahian et al. combine active vision SLAM with multi-UAV cooperative navigation to realize efficient exploration in unknown outdoor environment [27]. Xue et al. constructed the trajectory topology diagram based on SLAM and introduced the reinforcement learning local planner [28]. Wang et al. studied the collaborative optimization method of SLAM and trajectory optimization for complex patrol scenes [29]. This study verifies the guiding value of semantic map to trajectory generation. Fusic and Sitharthan applied the adaptive particle swarm optimization algorithm to the trajectory optimization of mobile robots based on 2D lidar [30].

Current research faces three major challenges: first, existing methods predominantly rely on single-scale semantics, making it difficult to jointly model relationships across pixel, object, and scene levels; second, semantic and geometric modules are often statically coupled and unable to dynamically adjust according to environmental uncertainty; third, system robustness remains insufficient under extreme lighting, unknown objects, and limited computational resources. Future efforts should focus on breaking through these limitations by: constructing temporal multi-scale semantic representations to enhance multi-level semantic collaboration, designing adaptive fusion mechanisms to achieve dynamic trade-offs between semantic and geometric cues, and integrating semantic priors with multi-sensor data to improve adaptability in open and dynamic environments.

Visual SLAM is undergoing a critical transformation from geometric drive to semantic enhancement. By deeply embedding multi-scale semantic constraints into the entire process of pose estimation, trajectory optimization, and error correction, the new generation SLAM system demonstrates advantages in dynamic interference suppression, trajectory rationality, and long-term stability. With the continuous advancement of lightweight semantic perception and adaptive optimization technology, semantic driven SLAM is expected to achieve large-scale implementation in practical scenarios such as service robots and autonomous driving.

3 SLAM algorithm model under multi-scale semantic constraints

The SLAM algorithm model proposed in this article under multi-scale semantic constraints aims to improve the rationality of trajectory optimization and error correction accuracy, and constructs a unified optimization framework that integrates pixel level, object level, and scene level semantic information. This framework abandons the loosely coupled design of semantic modules in traditional SLAM, which only serve as post-processing or auxiliary annotation. It deeply embeds multi-scale semantic priors into the entire chain of front-end tracking, keyframe management, pose graph optimization, and trajectory generation, forming a semantic geometric joint inference mechanism. The overall system adopts a three thread collaborative architecture (Figure 1). The tracking thread implements real-time pose estimation guided by semantics, the local mapping thread completes incremental construction of semantically stable keyframes, and the loop detection thread triggers global correction through semantic topology consistency verification.

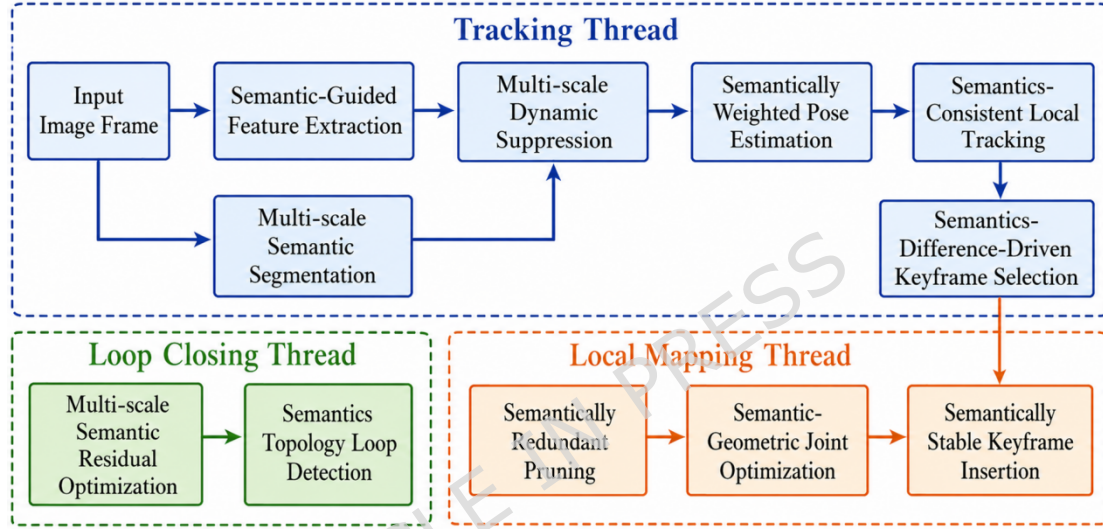


Figure 1 Architecture of SLAM system under multi-scale semantic constraints

The system first extracts multi-level semantic representations of the input image in real-time through a lightweight multi branch semantic segmentation network. The low-level branch outputs a pixel level semantic mask $s_p(x)$, which is used to identify basic semantic regions such as floors, walls, and dynamic objects

$$s_p(x) \in \{0,1\}^C \quad (1)$$

Where C is the number of semantic categories. The mid-level branch generates object level bounding boxes and category labels based on instance segmentation, with the semantic confidence of the i -th object denoted as $\phi_i \in [0,1]$. Its 3D center position is $o_i \in \mathbb{R}^3$.

The network is based on MobileNetV3, and it is divided into three branches after the third inverse residual block, sharing the early convolution features to reduce redundancy. Pixel-level branches retain high-resolution feature maps, object-level branches introduce attention mechanism to extract instance features, and scene-level branches use global average pooling to reduce dimensions. The budget allocation of calculation is: about 60% at pixel level, about 30% at object level and about 10% at scene level, so as to strike a balance between accuracy and real-time performance.

The high-level branch infers the room function type or road topology structure through the scene parsing module, and outputs a scene level semantic vector $s_g \in \mathbb{R}^K$. After alignment and filtering, the three types of semantic outputs together form a Multi scale Semantic Graph (MSG). The node set is defined as:

$$V = \{(T_k, s_p^k, \{o_i^k, \phi_i^k\}, s_g^k)\}_{k=1}^N \quad (2)$$

Among them, $T_k \in SE(3)$ is the camera pose at frame k .

The scene semantic vector extraction module uses global average pooling+two-layer fully connected to output 256 dimensions. During training, cross entropy and contrastive loss are used for joint optimization to enhance the compactness of features in similar scenarios. In closed-loop detection, the cosine similarity threshold is set to 0.85, achieving a balance between anti aliasing and maintaining high recall.

In the front-end tracking phase, the system adopts an improved direct sparse tracking strategy, but introduces pixel level semantic masks to weight and suppress the photometric error term. The photometric residual is defined as:

$$r_{\text{photo}}(x) = w(s_p(x)) \cdot |\mathcal{I}(\pi(TX)) - I_{\text{ref}}(x)| \quad (3)$$

$$w(s_p(x)) = 1 - \alpha \cdot I_{\text{dynamic}}(s_p(x)) \quad (4)$$

Among them, $\alpha \in [0, 1]$ is the dynamic suppression coefficient, and I_{dynamic} is the dynamic region indicator function. At the same time, object level semantic guidance is used to sample feature points, and the probability of key point selection is:

$$P(x) \propto \phi(x) \cdot \|\nabla \mathcal{I}(x)\|_2 \quad (5)$$

Ensure sampling is prioritized in areas of high semantic confidence and geometric salience. This significantly enhances the reliability of feature matching, particularly in scenes with repetitive structures or weak textures.

The keyframe selection mechanism has also been semantically reconstructed. Traditional methods usually trigger keyframe insertion based on motion amplitude or disparity threshold, which can easily generate a large number of redundant frames in semantically redundant areas, and may miss key information at semantic turning points [31]. Keyframe selection is based on MSG structural difference measurement:

$$\Delta_{\text{sem}} = \lambda_1 \cdot d_{\text{obj}} + \lambda_2 \cdot d_{\text{scene}} \quad (6)$$

$$d_{\text{obj}} = \frac{1}{M} \sum_{i=1}^M \|o_i^{\text{curr}} - o_i^{\text{last}}\|_2 \quad (7)$$

$$d_{\text{scene}} = 1 - \cos(s_g^{\text{curr}}, s_g^{\text{last}}) \quad (8)$$

If $\Delta_{\text{sem}} > \tau$, a new key frame is inserted. This mechanism optimizes the sparsity of the map and provides a structured semantic node for the subsequent trajectory optimization.

The backend optimization module constitutes the core innovation of this model. Backend optimization to build a hybrid pose graph, with a total energy function of:

$$E_{\text{total}} = \underbrace{\sum_{(i,j) \in \mathcal{E}_g} \|r_g(T_i, T_j)\|_{\Sigma_g}^2}_{\text{Geometric term}} + \underbrace{\sum_{(i,j) \in \mathcal{E}_s} \omega_{ij} \|r_s(T_i, T_j)\|_{\Sigma_g}^2}_{\text{Semantic item}} \quad (9)$$

The semantic residual r_s is defined as the object position consistency error:

$$r_s(T_i, T_j) = R_i^{-1}(o_k - t_i) - R_j^{-1}(o_k - t_j) \quad (10)$$

$$\omega_{ij} = \sigma(\phi_k) \cdot (1 - \beta \cdot \text{Var}(X_k)) \quad (11)$$

$\sigma(\cdot)$ is the Sigmoid function, and β is the geometric uncertainty adjustment factor. For co viewed object instances, if their 3D reconstruction positions in different keyframes conflict with semantic categories, a semantic consistency penalty term is introduced; For scene level semantics, if there is an unreasonable jump in the scene category to which consecutive keyframes belong, topological smoothing constraints are applied.

When the same object instance is observed in a new keyframe, the system uses the inverse variance weighted average method to update its position mean and variance. The weight of the new observation is inversely proportional to its geometric uncertainty. In long-term operation, the system only maintains uncertainty estimates for a limited number (usually ≤ 50) of object instances within the active keyframe window. This sparse maintenance strategy ensures low memory usage, a single update

computational complexity of $O(1)$, and no significant overhead.

At the trajectory optimization level, the system utilizes the MSG to construct a semantic topological representation. The nodes represent semantically stable keyframes, and the edges represent the semantic connectivity of passable areas. During pose graph optimization, the estimated camera trajectory is constrained to follow the principle of semantic rationality. This means the optimization process prioritizes poses that align with traversable regions, avoids areas with dense dynamic objects, and maintains smooth transitions at semantic boundaries. The edge weights in this semantic topological graph are defined as:

$$w_{\text{nav}}(u, v) = \gamma \text{IoU}(s_p^u, s_p^v) + (1 - \gamma) \cdot \exp(-\|T_u - T_v\|) \quad (12)$$

The guiding trajectory is generated along areas with high passability. This strategy ensures that the generated path aligns more closely with human spatial cognitive logic, while reducing positioning failures caused by traversing unstructured areas.

The error correction mechanism is implemented through multi-scale semantic residual feedback [32]. In error correction, local beam adjustment freezes the depth of semantically stable regions:

$$\min_{T, D_{\text{unstable}}} \sum_{x \notin \Omega_{\text{static}}} r_{\text{photo}}(x)^2 \quad (13)$$

$$\Omega_{\text{static}} = \{x | s_p(x) = \text{"wall" or "floor"}\} \quad (14)$$

Global closed-loop verification must meet the following requirements:

$$\|s_g^i - s_g^j\|_2 < \epsilon_{\text{scene}} \quad \text{and} \quad \text{Hungarian}(\mathcal{O}^i, \mathcal{O}^j) < \epsilon_{\text{obj}} \quad (15)$$

If this condition is not met, the closed loop is rejected. The whole algorithm is implemented in PyTorch framework, and the semantic segmentation module is based on MobileNetV3 backbone, which ensures the real-time reasoning ability on the embedded platform (average 28FPS on Jetson AGX Xavier) and meets the actual deployment requirements. The double correction mechanism effectively suppresses short-term disturbance and long-term drift, and significantly improves the stability of the system in long-term operation.

4 Experimental verification and performance assessment

To evaluate the effectiveness of the proposed algorithm in trajectory optimization and error correction, this article conducted systematic experiments on TUM RGB-D, KITTI, and a self collected urban street view dataset (Urban Campus-2025). All comparison methods are run on the same hardware platform and software environment to ensure fairness. The method used in this article is referred to as MS-SLAM (Multi scale Semantic SLAM), with baselines including ORB-SLAM3, DS-SLAM, SSM-SLAM, and VSO.

In order to ensure the practicality of the proposed method, a self collected dataset "City Campus -2025" was constructed. The data collection was conducted in different campus environments, including indoor corridors, outdoor sidewalks, and mixed traffic areas, covering a total sequence length of approximately 4.5 kilometers over an 8-hour operation. The sensor configuration includes an Intel RealSense D435i RGB-D camera and a built-in 6-degree-of-freedom IMU with a resolution of 640x480 and a frame rate of 30 Hz. For annotation, a semi-automatic pipeline is used. Generate initial semantic labels for 15 common categories using a pre trained segmentation model, followed by manual refinement to ensure high annotation accuracy.

Dynamic pedestrians frequently cross the field of view, which interferes with pose estimation. Figure 2 shows the temporal variation of ATE for each method. The MS-SLAM curve is generally flat, with a maximum ATE of 0.082 m, much lower than ORB-SLAM3 (0.193 m) and DS-SLAM (0.167 m). Especially in the interval of frames 1200-1750 (pedestrian dense area), the error amplification of MS-SLAM is only 0.011 m, while ORB-SLAM3 increases sharply by 0.063 m, indicating that multi-scale semantic constraints effectively suppress local drift caused by dynamic interference.

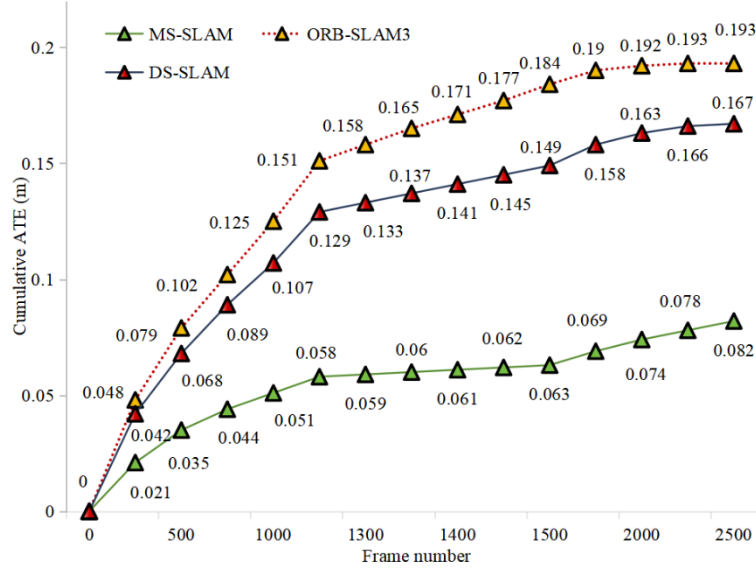


Figure 2 Curves of ATE varying with time for different methods on TUM sequences

In urban roads with a large number of repeated building facades, geometric features are prone to mismatches. Figure 3 shows the curve of RPE (translational component over 10m interval) as a function of mileage. The mean RPE of MS-SLAM is 0.018 m, with a standard deviation of 0.007 m; in contrast, VSO is 0.031 ± 0.015 m, and SSM-SLAM is 0.029 ± 0.013 m. It is worth noting that the RPE fluctuation amplitude of MS-SLAM is the smallest on the 3.2-3.8 km section. This validates the compensatory effect of object level semantics on weak texture regions.

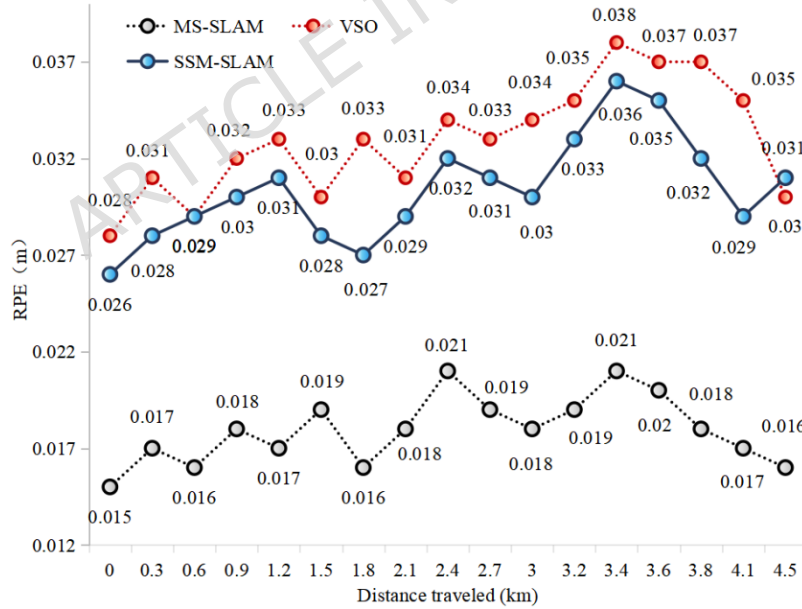


Figure 3 Trend of RPE varying with mileage on KITTI 00 sequence

To further verify the rationality of trajectory optimization, this article records the alignment between camera trajectories and semantic maps in the Urban Campus-2025 dataset. Figure 4 shows the curve of the distance (in meters) of the path deviation from the centerline of the passable area over time. The average deviation of MS-SLAM is only 0.24 m, while ORB-SLAM3 reaches 0.68 m, often crossing flower beds or step edges due to geometric optimality. This result directly

reflects the optimization effect of semantic topology guidance on trajectory generation.

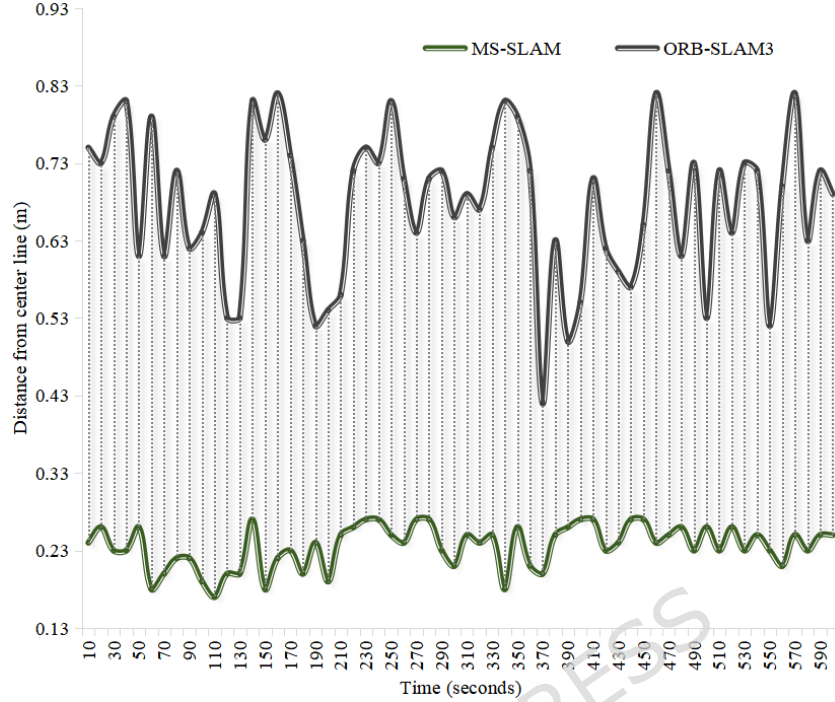


Figure 4 Distance of path deviation from passable centerline in Urban Campus-2025

In terms of error correction capability, Figure 5 compares the trend of ATE accumulation under long-term operation. The final ATE of MS-SLAM was 0.103 m, while all other methods exceeded 0.25 m. In the 25-30 minute interval, MS-SLAM successfully triggered global correction through scene level semantic closure, and the ATE decreased by 0.031 m. However, DS-SLAM detected geometric loops but mistakenly identified similar offices as the same location due to unverified semantic consistency, resulting in correction failure and an ATE increase of 0.018 m.

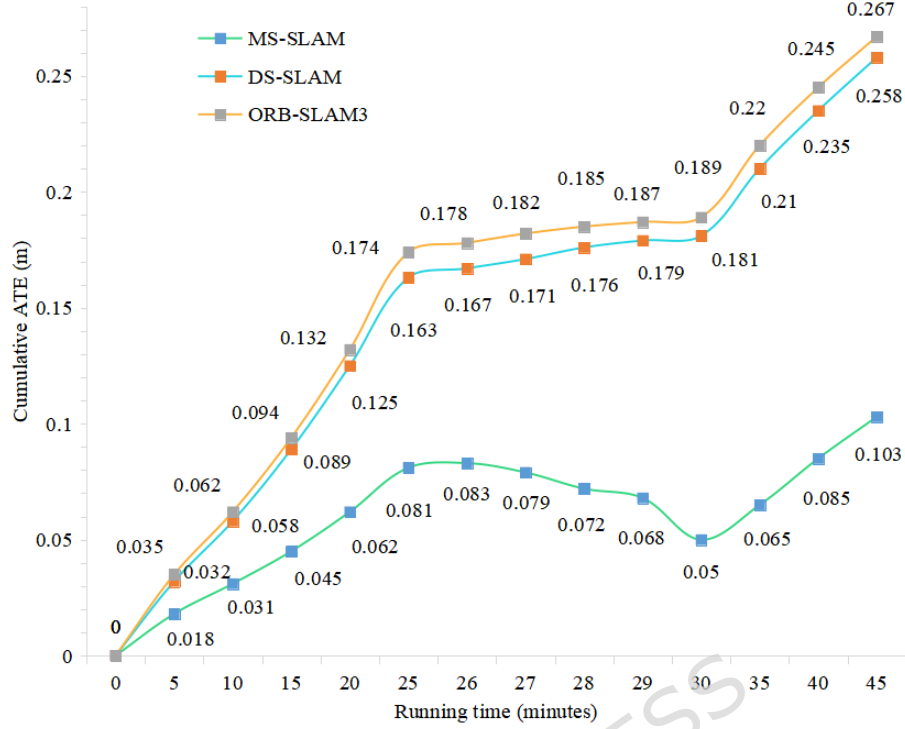


Figure 5 ATE accumulation and correction effect under long-term operation

Computational efficiency is a key indicator of real-time SLAM. Figure 6 shows the distribution of single frame processing delay (in milliseconds) for each method on Urban Campus-2025. The mean of MS-SLAM is 35.2 ms (≈ 28.4 FPS), slightly higher than ORB-SLAM3 (28.7 ms), but better than DS-SLAM (52.1 ms) and VSO (48.6 ms). The increase in latency is mainly due to the semantic segmentation module, but it is controlled within an acceptable range through lightweight design.

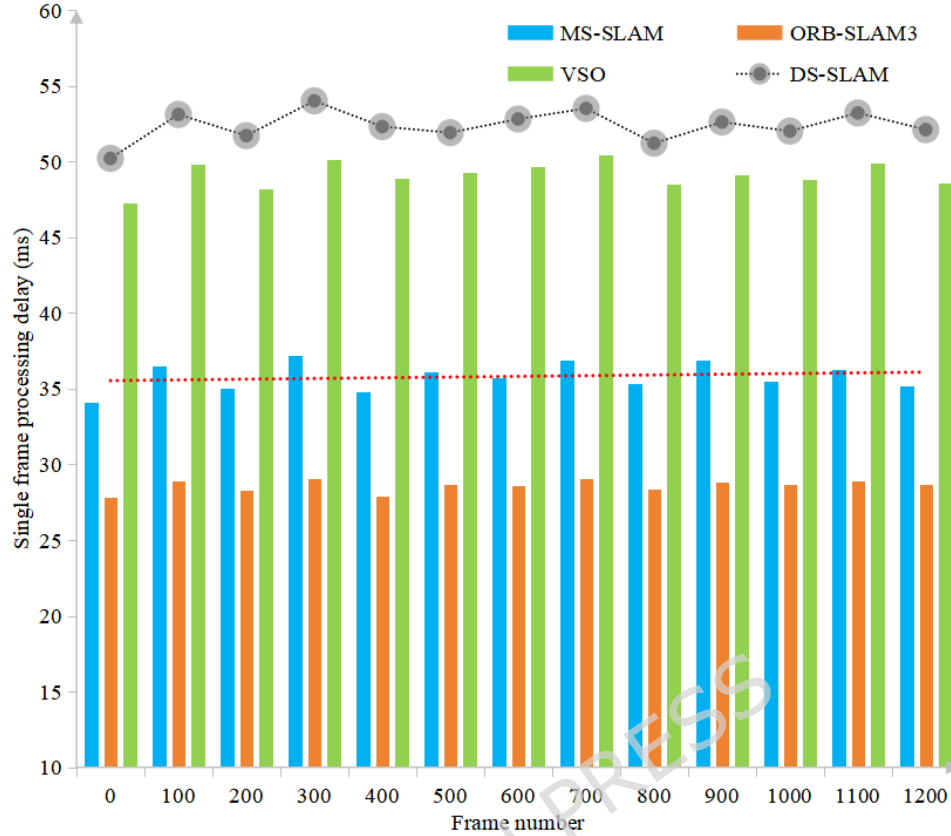


Figure 6 Comparison of single-frame processing latency distribution

Figure 7 analyzes the contribution of semantic confidence to pose correction. The data shows that when the confidence level is greater than 0.85, the correction amount is concentrated in the range of 0.02-0.05 m and is negatively correlated with geometric residuals ($r=-0.73$), indicating that high confidence semantics effectively guide the direction of error correction; In low confidence regions (<0.6), the correction amount is dispersed and the amplitude is small, and the system automatically reduces its weight to avoid noise interference.

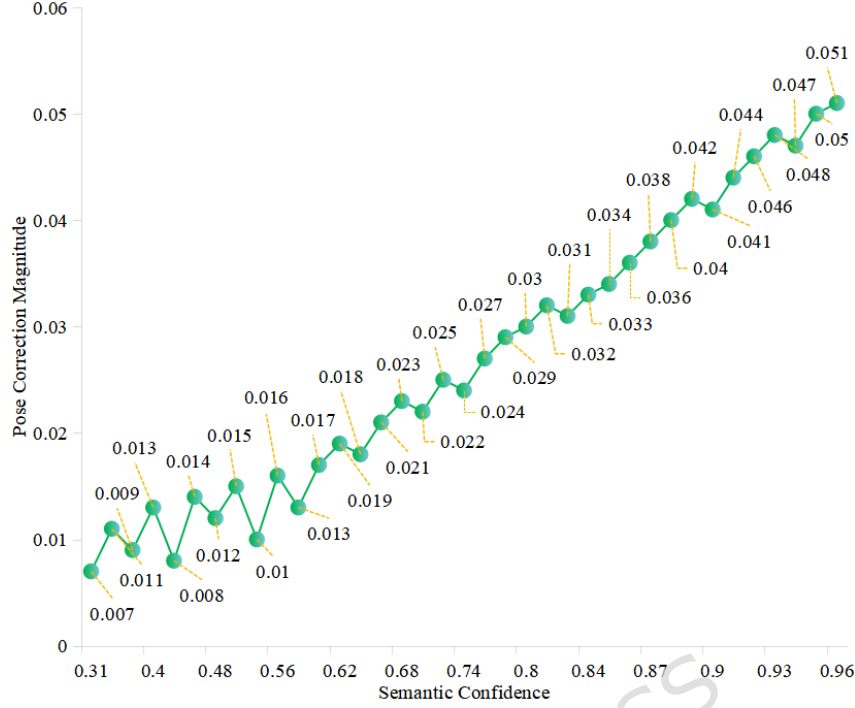


Figure 7 Relationship between semantic confidence and pose correction amount

Table 1 quantitatively evaluates overall performance from four dimensions. ATE (m), RPE (m), Semantic annotation IoU (%) and memory usage (MB). MS-SLAM is optimal on both ATE and RPE (0.112 m, 0.021 m), with a semantic IoU of 78.4%, which is higher than DS-SLAM with only additional semantic modules (62.1%), demonstrating the effectiveness of its semantic geometric joint training; The memory usage is 412 MB, slightly higher than ORB-SLAM3 (328 MB), but much lower than the dense semantic method VSO (687 MB). The data in the table further confirms that multi-scale semantic constraints synchronously improve positioning accuracy, semantic quality, and system compactness without significantly sacrificing efficiency.

Table 1 Performance Comparison of SLAM Methods

Method	ATE (m)	RPE (m)	Semantic IoU (%)	Memory Usage (MB)
ORB-SLAM3	0.237	0.048	-	328
DS-SLAM	0.205	0.041	62.1	503
SSM-SLAM	0.189	0.036	67.3	476
VSO	0.163	0.031	71.8	687
MS-SLAM	0.112	0.021	78.4	412

The experimental results show that MS-SLAM achieves a good balance between dynamic interference suppression, repetitive structure identification, path semantic rationality, long-term drift control, and real-time performance. Its core advantages stem from the deep coupling of multi-scale semantic information - pixel level masking filters dynamic noise, object level instances provide structural anchor points, scene level topology ensures global consistency, and the three work together to drive trajectory optimization and error correction, forming a closed-loop enhancement mechanism.

5 System implementation and deployment analysis

To verify the engineering feasibility of the proposed MS-SLAM system, this article constructs a modular software architecture based on ROS 2 Humble and completes end-to-end deployment on the Jetson AGX Xavier embedded platform through system

implementation and deployment analysis. The system adopts an asynchronous pipeline design: the semantic segmentation module runs at 15 Hz on the Tensor RT acceleration engine, outputting multi-scale semantic graphs; The geometric tracker processes sparse features and photometric residuals at 30 Hz; The backend optimizer triggers when a keyframe or closed loop is detected, and uses the g2o solver for incremental pose graph optimization. Each module communicates through shared memory and zero copy message mechanism, effectively reducing latency. **In order to match the frequency difference between 15 Hz semantic segmentation and 30 Hz geometric tracking, the system adopts asynchronous cache and nearest neighbor interpolation: the geometric thread runs continuously at 30 Hz, and the semantic mask of the nearest cache is taken according to the timestamp; Non-key frames reuse the semantic topology of the previous key frame, and complete semantic reasoning is performed only when the key frame is inserted or closed-loop.**

Table 2 shows the average resource consumption of the system during 30 minutes of continuous operation in the Urban Campus-2025 sequence. The semantic front-end occupies 1.8 GB of GPU memory and an average of 1.2 CPU cores; The geometric tracker occupies 0.7 cores; The backend optimization peak occupies 2.3 cores, but on average only 0.4 cores. The overall memory usage remains stable at 412 MB without any significant leaks, meeting the long-term operational requirements of embedded platforms.

Table 2 Resource Usage Analysis of Each MS-SLAM Module

Module	CPU Cores (Average)	GPU VRAM (GB)	Memory (MB)
Semantic Frontend	1.2	1.8	186
Geometric Tracker	0.7	0.0	98
Backend Optimizer	0.4	0.0	128
Total	2.3	1.8	412

In a real campus inspection task, the system works continuously for 8 hours and records its operational stability. Table 3 shows that the average frame rate remains at 27.6 FPS with a standard deviation of only 1.3; A total of 127 semantic loops were triggered, of which 121 passed topology consistency verification, with a loop acceptance rate of 95.3%. During this period, there were three brief tracking losses, and the system recovered within 2 seconds through semantic relocation without causing any interruption in mapping. This result indicates that multi-scale semantics not only improves accuracy, but also enhances the system's fault tolerance and self recovery capabilities in real-world environments.

Table 3 Performance Stability Statistics for Long-Term Operation

Indicator	Value
Average Frame Rate (FPS)	27.6 ± 1.3
Number of Semantic Loop Closure Triggers	127
Loop Closure Acceptance Rate	95.3%
Number of Tracking Losses	3
Average Recovery Time (s)	1.8

In order to evaluate the contribution of each semantic level, ablation research is carried out by removing specific semantic constraints in turn. As shown in Table 4, only the geometric baseline system shows the highest error and failure rate. The introduction of pixel-level semantics mainly reduces the tracking failure rate by filtering dynamic noise. By providing stable structural anchors in weak texture areas, the consistency of local trajectories is improved by increasing object-level semantics. Scene-level semantics further enhances global consistency and prevents large-scale drift during long-term operation. The complete MS-SLAM configuration integrates all three scales, achieving the best balance between positioning accuracy and trajectory smoothness.

Table 4 Ablation study on the contribution of multi-scale semantic constraints

Configuration	ATE (m)	RPE (m)	Trajectory Smoothness	Tracking Failure Rate
Baseline (Geometric only)	0.245	0.052	1.00 (Ref)	4.5
+ Pixel-level semantics	0.198	0.043	1.12	3.1
+ Object-level semantics	0.165	0.035	1.18	2.2
+ Scene-level semantics	0.151	0.038	1.15	2.0
Full MS-SLAM (All scales)	0.112	0.021	1.31	1.8

6 Discussion

Existing semantic SLAM systems often regard semantic modules as post-processing add-ons, which are loosely coupled with front-end tracking, trajectory optimization and error correction processes, making it difficult to achieve collaborative optimization under real-time constraints [33]. Especially in the embedded platform with limited resources, the high-cost semantic segmentation network further aggravates the contradiction between accuracy and efficiency [34,35]. The multi-scale semantic constraint mechanism proposed in this article shows significant advantages in improving the robustness of SLAM system, but its performance boundary is still subject to the reliability of semantic perception and the applicability of environmental priors. Experiments show that the dependence of the system on semantic constraints may lead to bias when there are a large number of unknown object categories in the scene or extreme lighting conditions lead to a sudden drop in confidence in semantic segmentation. This suggests that in the future, it is necessary to introduce an uncertain perceptual mechanism to dynamically adjust the fusion ratio of semantic and geometric constraints, rather than adopting a fixed weighting strategy. The construction of multi-scale semantics itself also brings about an increase in the complexity of calculation and modeling [36]. Although this article controls inference latency through lightweight networks, semantic modules may still become bottlenecks on embedded platforms with extremely limited resources. In addition, the current framework assumes natural alignment between semantic levels, ignoring the cumulative effect of pixel level noise propagating to higher-level semantics. In practice, it has been observed that when there is boundary jitter in instance segmentation, the object level topology map will be frequently reconstructed, resulting in unstable keyframe selection. This issue is particularly prominent in fast motion scenarios, indicating that the temporal consistency of multi-scale representations has not been fully guaranteed.

In the complex and changeable dynamic environment, the system relies on semantic features, which may cause its performance to decline when facing unknown objects or under extreme lighting conditions. In order to enhance the robustness of the system in the scene of feature degradation, the cross-domain adaptive feature processing mechanism can be considered in future research. For example, the adaptive decomposition network is used for reference to strengthen the feature extraction ability in complex environment [37]. Multi-channel complex residual network can also be used to optimize the adaptability of decision boundary [38]. As far

as feature matching and alignment in outdoor scenes are concerned, the combination of robust three-stage registration strategy can also help to alleviate the error accumulation caused by perceptual aliasing to some extent [39].

From the perspective of trajectory optimization, although semantic guidance makes trajectories more in line with human spatial cognition, it may also sacrifice local geometric optimality. This reflects the inherent tension between semantic rationality and geometric accuracy, which needs to be explicitly modeled in the objective function to balance the two, rather than a one-way constraint. At a deeper level, current semantic constraints still rely on the static knowledge base of pre trained models, lacking adaptability to dynamic environmental evolution. In the real world, the functionality of the same space may change over time, and fixed semantic labels cannot capture such changes. If the SLAM system is deployed in long-term operational scenarios, online semantic learning or incremental scene understanding modules must be introduced, otherwise semantic priors will gradually become ineffective. Although this experiment covers various scenarios such as indoor, urban roads, and campuses, it does not involve extreme dynamic environments. In such scenarios, even if dynamic regions are removed, the remaining static structure may not be sufficient to support reliable positioning. Future work can explore the deep fusion of semantic and inertial, laser and other heterogeneous sensors to compensate for the information loss of pure vision under extreme conditions. Overall, multi-scale semantics provide new optimization dimensions for SLAM, but its true implementation still needs to address the dynamic adaptation and robust fusion issues in the perception decision correction loop.

7 Conclusions

This article focuses on the accuracy and stability bottlenecks of visual SLAM in complex dynamic environments, and constructs a real-time trajectory optimization and error correction technology system with multi-scale semantic constraints as the core. By deeply integrating pixel level semantic masks, object level instance layouts, and scene level topological structures, the system achieves effective filtering of dynamic interference and introduces semantic rationality criteria that conform to human spatial cognition at the trajectory generation level. The experimental results validate the comprehensive advantages of this framework: on multi-source datasets from indoor office, urban road, and campus scenes, MS-SLAM achieves an average ATE of 0.112 meters, outperforming ORB-SLAM3 (0.237 meters) and DS-SLAM (0.205 meters); its RPE is 0.021 meters, indicating stronger local motion consistency; the semantic annotation IoU reaches 78.4%, confirming that joint semantic-geometric training enhances semantic quality; meanwhile, a single-frame latency of 35.2 milliseconds ensures real-time performance on embedded platforms.

During a continuous 45 minute test run, MS-SLAM successfully triggered global calibration through scene level semantic closure, resulting in an ATE of only 0.103 meters. The comparison method generally exceeds 0.25 meters, highlighting its drift suppression capability in long-term deployment. In terms of trajectory optimization, the system deviates from the centerline of the passable area by an average of only 0.24 meters in Urban Campus-2025, which is much better than the pure geometric method of 0.68 meters. This indicates that semantic topology guidance effectively avoids unstructured traversal behavior.

However, this study still has limitations. The semantic performance is highly dependent on the generalization ability of the segmentation model and may degrade in the absence of categories or extreme lighting conditions; The temporal consistency of multi-scale representation also needs to be further strengthened. Future work will focus on online semantic adaptation, uncertainty perception fusion, and multimodal sensor collaboration to promote the migration of this technology to more open and dynamic real-world scenarios.

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Conflict of interests

Authors declare that there is no any competing interest.

Data available statement

Data can made available from the corresponding author upon reasonable reason.

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